25901073 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCAQC202 - REPRODUCTIVE PHYSIOLOGY, ENDOCRINOLOGY AND INDUCED BREEDING TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer questions

Answer all questions

Each question carries **1** marks

1. Mention one advantage of sex reversal in fishes. [K] / [CO1]
2. Mention the sexual dimorphism between male and female *Amia calva*. [An] / [CO1]
3. What is meant by genetic sex determination ? [K] / [CO1]
4. What is Gonadotropin Releasing Inhibitory Factor? [U] / [CO2]
5. Mention the hormones involved in the migration and concentration of pigments in epidermal chromatophores and Retinal cells in Crustaceans? [S] / [CO3]
6. Outline the functions of GTH 1 and GTH 2 [U] / [CO2]
7. Name the hormone present in eye stalk that inhibits maturity and spawning in crustaceans. [K] / [CO4]
8. Name any two artificial hormones used in induced breeding of fin fishes. [K] / [CO4]

9. What is the method of preservation of biological material at very low temperature in liquid nitrogen? [U] / [CO5]

10. What is invagination? [U] / [CO5]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

11. Descibe sexual dimorphism in crabs. [U] / [CO1]

12. Explain male reproductive system in prawn with suitable diagram. [K] / [CO1]

13. Describe oviparity, viviparity and ovoviviparity in fishes. [K] / [CO1]

14. Discuss the position, and functions of the Thyroid gland, Ultimobranchial glands, Pineal organ , Chromaffin tissue and Pancreatic islets in Teleosts. [U] / [CO2]

15. Explain the position, and functions of the Y-organs and Mandibular organs in Crustacea. [S] / [CO3]

16. Describe the advantages of producing monosex populations in aquaculture. [K] / [CO4]

17. Describe the procedure of hypophysation. [K] / [CO4]

18. Describe the gonadal level of control in induced maturation and spawning. [K] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

19. Explain the maturity stages of male and female reproductive organs in teleosts with diagrams. [U] / [CO1]

20. Summarise the different hormones produced by the Pituitary and Urophysis in fishes and explain their functions. [U] / [CO2]

21. Describe anaesthetics used in fish breeding and handling, their importance and methods of administering anesthetics. [K] / [CO4]

[10x1 = 10]